

Machine learning-based outlier detection for pipeline in-line inspection data

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ARTICLE INFO

Keywords:

Data quality
Outlier detection
Machine learning
In-line inspection (ILI)
Statistical analysis

ABSTRACT

Pipeline companies are facing challenges in maintaining the integrity and reliability of their pipelines. They are working towards predictive maintenance using machine learning-based approaches to predicting anomalies. Training machine learning models requires sufficient data. Data quality is therefore becoming important because inaccurate data will lead to an inaccurate or wrong decision on pipeline condition assessment and the following management. This research paper intends to address the data quality issues of pipeline inspection data such as in-line inspection (ILI) data using machine learning models. Different machine learning models developed by random forest regression, linear regression, and nearest neighbors' methods were tested to detect outliers in the ILI data. In this paper, the ILI data collected from an oil pipeline over a period of 22 years was applied to testing and analysis. To verify the outlier detection results of machine learning models, we used statistical analysis including Z-score method to check and find if there are any gaps in the analysis. It verifies that all these methods show almost the same or very similar results for the detection of the outliers. Hence, this study presents a robust method for the field applications in the pipeline industry.

1. Introduction

Energy pipelines play an important role in transporting commodities, including gas, oil, and other supplies to customers around the world [1, 2]. With the passage of time and age, these pipelines deteriorate and there is an increasing chance of failure, which can have catastrophic consequences for the economy, public safety, the environment, business interruptions, and maintenance and repair costs [3–5]. Pipelines that have corroded are prone to deteriorate in a way that is influenced by the local condition, either on the inner or outside wall, resulting in a loss of metal [6,7]. The various soil compositions surrounding the pipeline, the method of its construction, its purpose, workmanship, materials selection, and upkeep, all contribute to pipeline deterioration. For instance, numerous studies have observed that soils containing elevated concentrations of chlorides, sulfates, and acidic pH, as well as the existence of bacteria, fungi, or algae, significantly affect the integrity of the pipe wall [8–11]. The integrity of these pipelines is crucial to avoiding any operational failures and safety breaches [12,13].

To maintain pipeline integrity and reliability, pipeline companies are working with academic researchers to develop and verify machine learning and statistical models to predict anomalies for operating their

pipelines safely; however, using advanced analytics tools requires a good quality of data to achieve accurate results to assist in the decision-making process. In recent decades, the fast advancement of computer technology has led to increased interest in numerical simulation and machine learning approaches. These methods are capable of providing reliable probability estimates for complex issues in oil and gas pipelines [14,15]. Data quality issues can significantly impact the accuracy and reliability of overall pipeline integrity and condition assessment [16, 17]. Data which is collected from different systems often encounter issues, for instance, inadequate or missing data, which may require more information regarding the pipeline's history, maintenance records, and operational and process conditions [18]. Lacking a comprehensive dataset poses difficulties in making the right decisions regarding the pipeline's condition and successfully prioritizing repair or replacement activities [5].

In-line inspection (ILI) is one of the techniques used to identify and quantify corrosion issues in energy pipelines. ILI is used as a screening tool to assess the condition or detect defects in the pipeline periodically, for instance, every 2 to 5 years. The results of this examination consist of a few items below and illustrated in Fig. 1 [8].

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<https://doi.org/10.1016/j.ress.2024.110553>

Received 10 May 2024; Received in revised form 13 September 2024; Accepted 4 October 2024

Available online 12 October 2024

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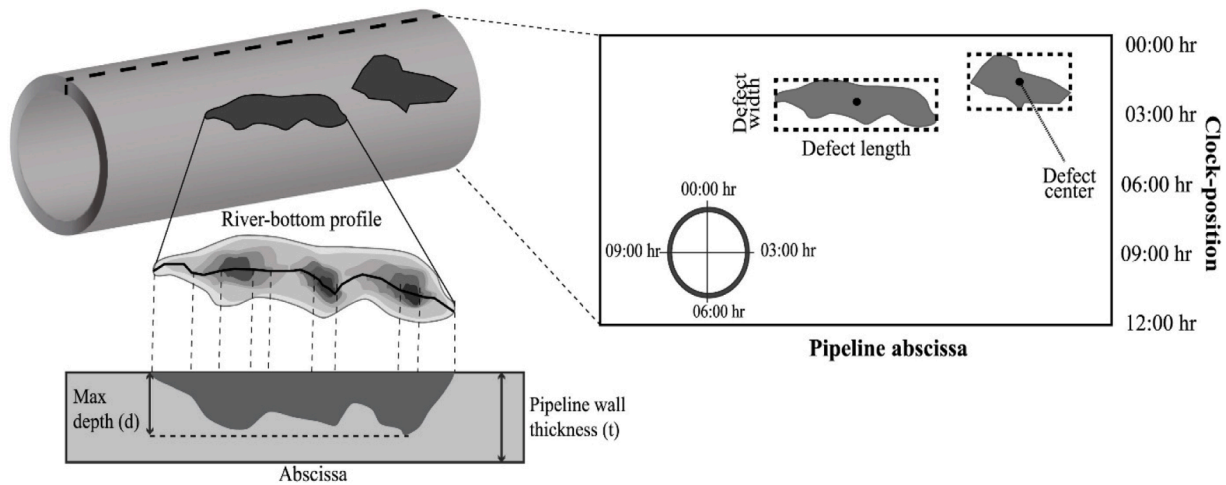


Fig. 1. Scheme of the dimensions of corrosion defects [8].

- (i) The dimensions of each anomaly (i.e., corrosion defect), including its depth, length, and width, as identified by the inspection tool. To achieve a detection, the anomaly depth must exceed a certain threshold, often set at 10 % of the pipe wall thickness.
- (ii) The precise position of each anomaly is determined by its distance down the pipe and its orientation, which is described using a clock-position analogy.
- (iii) The ILI results also encompass pipeline structural characteristics (e.g., diameter, wall thickness), pipeline distribution (e.g., inclination, latitude/longitude, altitude), and operating parameters such as the maximum operating pressure [10].

ILI employs various technologies, including intelligent pigs equipped with sensors that travel through pipelines to collect data on the condition of the pipeline walls. These pigs utilize technologies such as magnetic flux leakage (MFL), ultrasonic testing (UT), electromagnetic acoustic transducer (EMAT), and inertial measurement units (IMU) to evaluate pipeline integrity [19–22].

The data collected during ILI is vital for evaluating the remaining life of corroded pipelines, predicting corrosion growth rates, and identifying corrosion anomalies. By analyzing ILI data, pipeline operators can make informed decisions regarding maintenance, repair, and integrity management strategies. Repeat ILI runs offer insights into the growth of corrosion anomalies over time, enabling the estimation of corrosion rates and the prediction of future severity [23–25]. The processing of ILI data is a crucial step in assessing pipeline conditions and determining appropriate actions to ensure safe pipeline operation [26,27].

A significant challenge in ILI is accurately identifying and characterizing complex corrosion features. Ongoing advancements in ILI technologies and methodologies aim to overcome technical limitations in detecting and characterizing critical corrosion anomalies. These advancements seek to improve the reliability and accuracy of ILI data interpretation, ultimately enhancing pipeline integrity management practices [28–30]. Moreover, the integration of machine learning and data analytics into ILI data has shown promise in enhancing corrosion growth assessment and supporting informed decision-making processes [31].

The validation and calibration of ILI technologies are essential to ensure their effectiveness in detecting defects such as stress corrosion cracking (SCC) and mechanical damage features. Studies have evaluated the performance of EMAT ILI technologies in managing SCC in natural gas pipelines, underscoring the importance of validation processes in upholding pipeline integrity [21,32]. The development of new ILI tools, like magnetic eddy current inspection tools, reflects ongoing efforts to

enhance ILI technologies for pipeline inspection and maintenance [33]. In pipeline integrity management, ILI data is utilized to conduct fitness-for-service assessments and remaining life analyses of detected defects. By estimating corrosion growth rates and predicting defect behavior over time, pipeline operators can implement proactive maintenance strategies to prevent failures due to corrosion. Integrating ILI data with finite element analysis methods further enhances the accuracy of pipeline defect evaluation and failure prediction, contributing to the overall safety and reliability of pipeline operations [34,35].

By leveraging outlier detection algorithms, pipeline operators can identify outliers, ensuring the accuracy and integrity of the data processed [36]. These methods can aid in the direct classification of pipe conditions without the need for specialized intervention, thereby streamlining the detection process [37]. The use of machine learning algorithms in outlier detection contributes to the identification of anomalies in pipeline systems, allowing for prompt follow-up actions by human experts [38].

Outlier detection methods have been shown to improve the performance of machine learning models, contributing to more accurate and reliable decision-making processes [39,40]. Automated outlier detection approaches have been developed to replace manual methods, offering a more efficient and less error-prone solution for identifying outliers in real data [41]. Early detection of outliers in wireless sensor networks has been highlighted as a means to conserve network resources and prevent unnecessary transmissions [42].

Outlier detection is a critical component in various industrial fields including data science. Machine learning techniques are commonly used for outlier detection due to their effectiveness in identifying anomalies in raw data [43]. Machine learning models trained on diverse datasets have shown improved accuracy and efficiency in detecting outliers [44].

The combination of supervised and unsupervised techniques in machine learning frameworks has demonstrated the potential to enhance outlier detection effectiveness [45]. Furthermore, ensemble algorithms like XGBOD have been introduced to improve the detection of outliers in various datasets [46]. These advanced algorithms and techniques contribute to the enhanced identification of anomalies within datasets, thereby aiding in outlier detection processes.

Moreover, outlier detection methods have found applications in various domains, including healthcare and IoT, underscoring their versatility and effectiveness in identifying anomalies [47,48]. The utilization of outlier detection techniques across diverse fields highlights their importance in addressing critical issues such as risk assessment, anomaly identification, and complex disease characterization [47].

The integration of acoustic emission data with machine learning algorithms is crucial for achieving rapid and accurate leak detection,

Table 1
Significance of Addressing Data Quality Gaps in ILI Data.

Challenge	Impact	Benefit of Improved Data Quality
Missing values	Underestimation or overestimation of defect severity [57]	More accurate and reliable condition assessments
Inconsistent measurements	Reduced confidence in overall data	Enhanced accuracy and reliability
Outliers	Misdiagnosis of defects or unnecessary repairs	Optimized inspection and maintenance strategies
Overall poor data quality	Inefficient resource allocation [58]	Data-driven decision-making for pipeline management

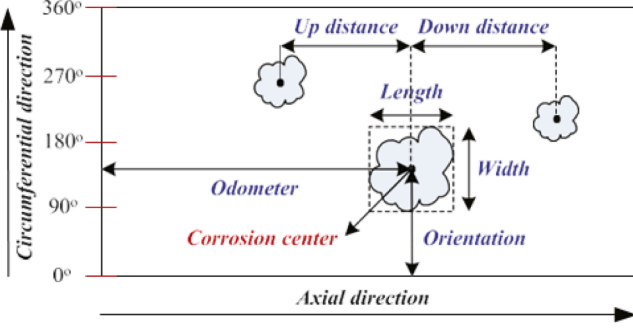


Fig. 2. Schematic of a corrosion feature on a 2D plane from one segment [59].

emphasizing the importance of real-world data integration with artificial intelligence [36]. Novel algorithms combining statistical methods and machine learning clustering techniques have been proposed to enhance outlier detection accuracy and control extreme values more effectively [40,49]. Machine learning-based anomaly detection methods offer flexibility, stability, and reliability, with a low probability of false alarms, ensuring robust outlier detection capabilities [50].

In practice, once ILI data is collected, it requires further analysis to understand pipeline's existing condition, identify anomalies, corrosion growth, and other defects. To achieve these objectives, a detailed analysis is required to understand the data, remove outliers, and make decisions that help pipeline owners in replacing or repairing the pipeline or its particular segment [5]. Inaccuracies in measurement, calibration, or recording might result in incorrect assessments of the reliability of the pipeline [51]. Inaccurate data can lead to incorrect maintenance decisions, which may result in unnecessary expenses or, more importantly, missing real threats to pipeline integrity [3,52].

This study explores data quality issues, and detection methods for outliers in raw ILI data. As per the authors' best knowledge, there is a lack of substantial analysis of ILI data outliers and discussing its overall impact on pipeline integrity management. Therefore, the authors of this paper focus on outliers' detection of ILI raw data using statistical and machine learning-based approaches. This work will provide valuable insights for researchers and pipeline owners, helping them appreciate the importance of data quality and its correlation with making more reliable and right decisions using advanced analytical techniques.

1.1. Background

The oil and gas industry is one of the most data-intensive sectors globally [53]. Oil and gas firms gather enormous volumes of data from multiple sources, including sensors on drilling rigs and pipelines, seismic surveys, and financial records [54,55]. This information may be utilized to save costs, increase safety and environmental performance, and improve the effectiveness of exploration and production [56].

Regular inspections and testing are crucial for ensuring the integrity

and reliability of energy pipelines. The in-line inspections (ILIs) are utilized to detect anomalies in oil and gas pipelines to assess their conditions. The ILI data is applied to prediction of anomalies and corrosion growth, which will assist pipeline operators in taking proactive action to avoid any unplanned downtimes and in scheduling the replacement of a part or a segment of the pipeline or considering other alternatives to prevent catastrophic failures.

The significance of addressing data quality gaps in ILI data is summarized in Table 1. The data quality issues may include missing values, inconsistent management, outliers and overall poor data quality. Outliers, representing anomalous behaviors, are extreme cases of class imbalance. Conventional methods struggle with class imbalance and outlier detection. Machine learning (ML) techniques promise to address these challenges by leveraging data-driven approaches.

1.2. Objective

The objective of this study is to discuss data quality issues and typically on detection of outliers. Various measurement and data processing factors can cause outliers in the ILI data set. Some common reasons may include ILI tool management errors, changes in pipeline operating conditions, pipeline conditions, pipeline features, data processing errors, localized conditions, noise and interference, out-of-spec conditions (operating conditions that fall outside the specified range for the ILI tool), human error in data interpretation or data entry, and natural variability (e.g., some outliers may arise due to natural variability or random fluctuations in the pipeline condition or measurement process). Addressing outliers in ILI data requires careful analysis and consideration of these factors to determine whether the measured values represent genuine anomalies in the pipeline or are due to measurement or processing errors. By giving priority to the elimination of outliers and establishing robust mechanisms for detecting outliers, pipeline owners can gain great benefits such as more accurate pipeline condition prediction, optimized maintenance schedules, reduced lifecycle costs, and enhanced pipeline integrity.

Therefore, this work proposes an outlier detection and identification method to ensure the detection results are reliable, and the process can be implemented in the industry practice.

1.3. Contribution

The main contributions of this work are concerned with the following three aspects:

- It proposes a method for outlier detection of energy pipeline ILI data by assessing commonly applied outlier detection techniques/methods;
- A new modification to the conventional interquartile range (IQR) method is proposed and the modified IQR method is certified to be effective for outlier detection of energy pipeline ILI data; and
- A general guideline for machine learning model selection for detection of outliers in ILI datasets is generated.

1.4. Limitation of the work

This work focuses on detecting outliers involved in ILI raw data by presenting a method to demonstrate the data quality concerns through machine learning and statistical analysis techniques within the field of pipeline condition assessment. In the analysis, only wall losses (%) data is utilized to demonstrate the analysis process. In the same way, data from other parameters can be tested. In this work, however, some other data like process, corrosion inhibitors, position, design data, etc. are not considered.

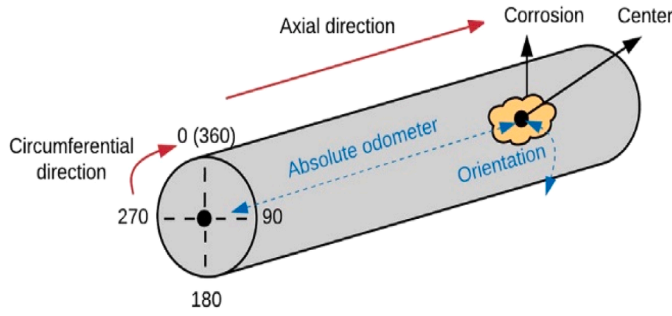


Fig. 3. Location of a corrosion defect on a pipeline [63].

2. Anomaly/defect matching using ILI data

2.1. Measurement of an anomaly/defect by an ILI tool

In essence, a pipeline can be seen as a three-dimensional longitudinal system, and it is reasonable to accurately employ models to simulate pipeline corrosion in three dimensions. Corrosion occurs exclusively within the pipe wall, which can be visualized as a two-dimensional manifold, as depicted in Fig. 2. The wall can be regarded as a 2D steel sheet curved into a cylindrical shape and fused along the lengthwise direction to form a pipe joint. Therefore, the corruptions are depicted on a two-dimensional flat surface, as illustrated in Fig. 2 [59].

The ILI data contains pre-processed and selected intrinsic features, which are listed below:

- **Odometer:** The Odometer measures the distance between the center point of this corrosion and the head or tail joint.
- **Orientation:** Orientation refers to the angle position in the circumferential direction, which is represented using a $0^\circ \sim 360^\circ$ angle analogy as illustrated in Fig. 2.
- **Length:** It is the horizontal span of a corrosion defect.
- **Width:** It is the vertical span of a corrosion defect.
- **Depth percentage:** It is the proportion of the depth of a corrosion defect to the wall thickness, the depth is measured from the flat surface of the pipe to the culet (the bottom of the corrosion defect), in the radial direction of the pipe.

These five attributes are the outcomes derived from the original ILI data [60–62]. Based on several manual matching methods, we posit that these variables are closely correlated with the location of the corrosion defect in both the reference and current ILIs. Two additional variables, "up corrosion distance" and "down corrosion distance," are derived from the raw ILI data using mathematical procedures to enhance the comprehensiveness and effectiveness of model training features.

The positions of anomalies determined from the ILI measurement correspond to the precise coordinates of the centre of the features as shown in Fig. 3. In feature matching procedures, a coordinate system consisting of two characteristics, namely "absolute odometer" and

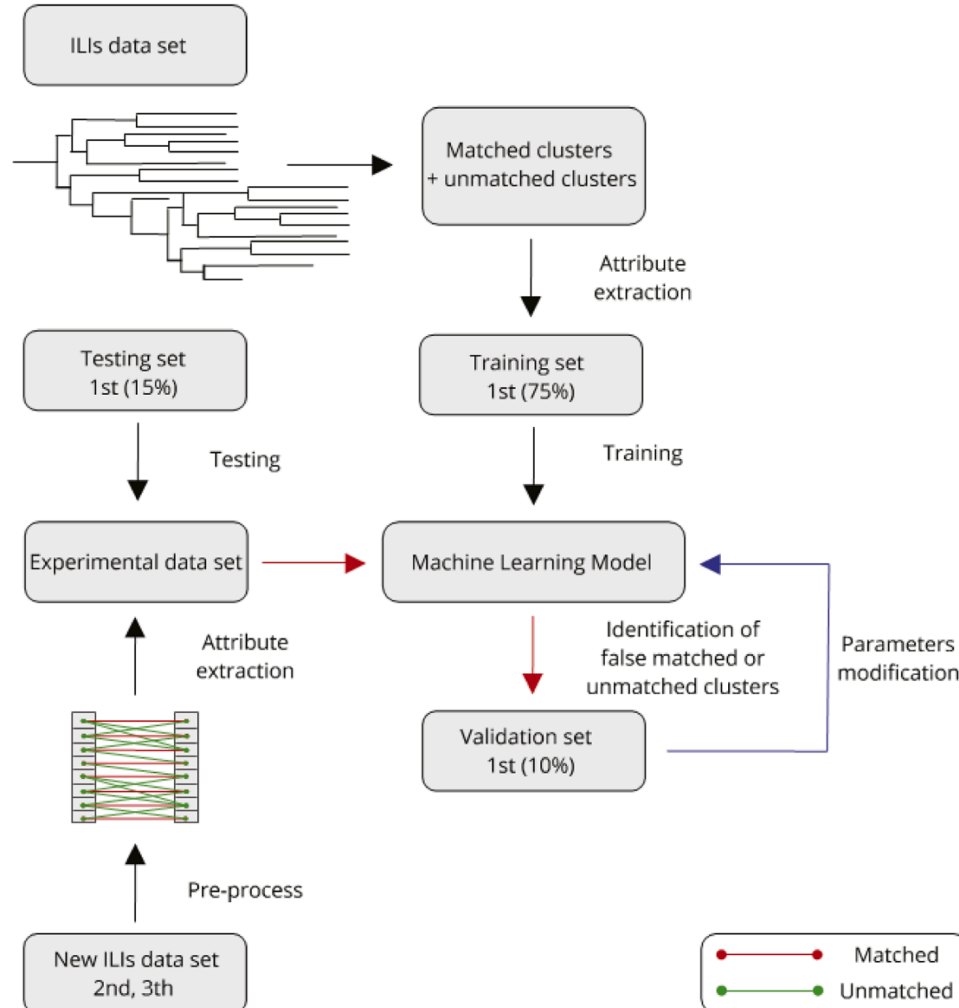
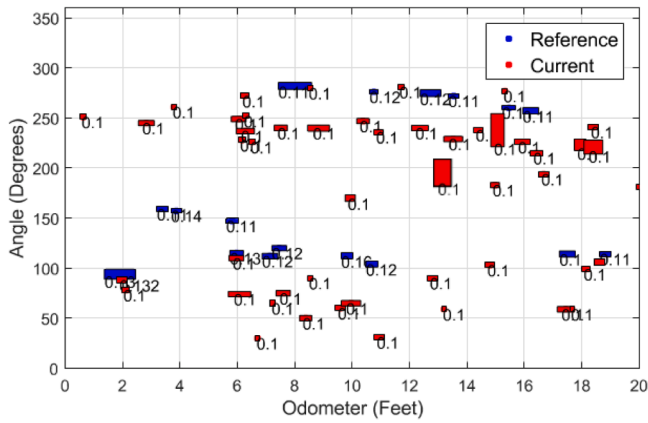


Fig. 4. A flowchart showing model training process [59].



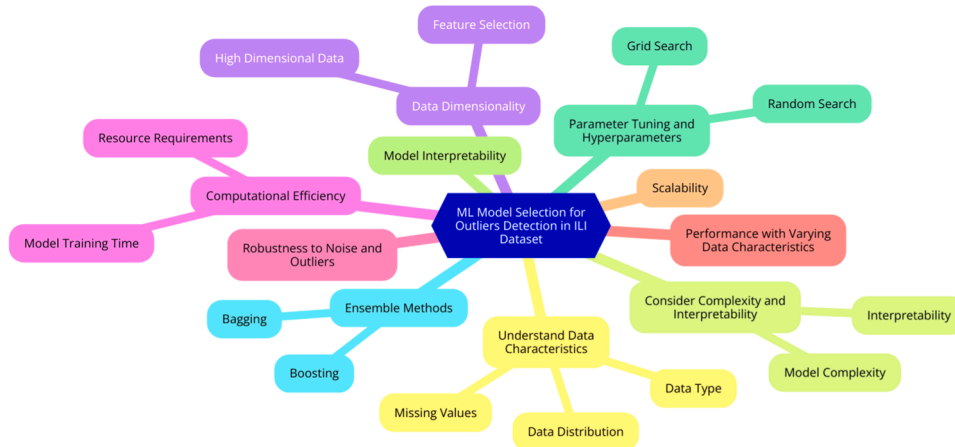


Fig. 7. Guideline of ML model selection for detection of outliers in ILI dataset.

corrosion features, which may reveal the remaining safe lifespan of each corrosion feature.

3. Machine learning models

3.1. Overview of machine learning models

The increasing use of machine-generated data in several fields, including energy pipelines, requires a strong framework to ensure data quality, especially in detecting outliers and identifying false data values in raw datasets collected from field inspection [66]. Machine learning models are powerful in identifying anomalies and outliers that could affect the accuracy of datasets [67–70].

The models were trained using a range of methodologies including unsupervised learning, supervised learning, statistical methods, ensemble approaches, hybrid techniques, and deep learning [71]. Understanding the strengths, limitations, and applications of these models is crucial for developing successful tactics to tackle data quality issues and improve the reliability of raw datasets [16]. Table 2 shows a few machine learning models to detect outliers and fake data in raw data.

3.2. Selection criteria for machine learning models

Choosing the most suitable machine learning model for identifying and eliminating outliers in pipeline ILI data is essential to achieve optimal outcomes to maintain pipeline integrity and reliability. A few key criteria for consideration are discussed below.

3.2.1. Data characteristics

- **Data size and dimensionality:** Working on a large ILI dataset, there are several features to be considered like defect size, length, depth, pipeline segment, location, orientation, etc. Keep in mind to select a model that is efficient for a specific data size and dimensionality.
- **Nature of data:** Consider whether the raw data is numerical, categorical, sequential, or a combination of all of them. Model selection needs to consider if a model is better suited to handle specific data types.
- **Distribution of outliers:** Examine the distribution of data points and assess whether it follows a normal distribution or exhibits skewed patterns, as this can impact the performance of certain models.

3.2.2. Training data size

- Assess the availability of a substantial amount of training data. Deep learning models often require large datasets for effective training, while simpler models may perform well with smaller datasets [79].

3.2.3. Detection goals

- **Sensitivity:** Consider the sensitivity of the chosen model to outliers. Some models, like Z-score and Mahalanobis Distance, are sensitive to outliers, while others, such as Isolation Forests, are designed to be robust against them [80].
- **Specificity:** How crucial is it to minimize false positives and accurately identify only true outliers? If specificity is paramount, models like interquartile range (IQR) or isolation forest might be suitable [81].

3.2.4. Computational sources

- **Computational Complexity:** Consider the computational resources available. Deep learning models can be computationally intensive, while simpler models may be more computationally efficient.

3.2.5. Performance evaluation

- **Evaluation metrics:** Define appropriate metrics for evaluating model performance, considering aspects such as precision, recall, F1-score, and area under the Receiver Operating Characteristic (ROC) curve depending on the specific goals of the analysis [82].
- **Baseline comparison:** Compare the performance of the chosen model with simpler baselines like statistical methods or domain-specific rules to assess its added value.

By carefully considering these criteria and conducting a thorough evaluation, pipeline operators can select the most effective machine learning model for addressing data quality gaps and enhancing the accuracy and reliability of pipeline condition assessments.

4. Guideline of ML model selection for detection of outliers in ILI dataset

Detecting outliers in pipeline ILI data is an important task. As discussed earlier, not all machine learning models are equally effective in detecting outliers in the ILI dataset. Hence, it is essential to choose the proper machine-learning technique based on the specific characteristics of ILI data and the goals of the outlier detection process. To select an

Table 3
ILI Probes and details.

ILI Probes	Details
Ultrasonic Testing (UT)	- This is the most widely used method for ILI due to its ability to accurately measure wall thickness and detect a variety of defects, including corrosion, pitting, and cracks [27]. - UT probes use sound waves to measure the time it takes for a signal to travel from the probe to the pipe wall and back. The thickness of the wall can be calculated based on this travel time [85]. - Different types of UT probes are available, such as thickness gauges, flaw detectors, and guided wave probes [86].
Magnetic Flux Leakage (MFL)	- This method uses magnets to create a magnetic field around the pipe. Any wall loss will disrupt the field, and this disruption can be detected by sensors on the probe. - MFL is good at detecting large areas of wall loss, but it can be less sensitive to smaller defects [87].
Eddy Current Testing (ECT)	- This method uses an electromagnetic field to induce eddy currents in the pipe wall. The strength and distribution of these currents are affected by the thickness and conductivity of the wall [88,89]. - ECT is good at detecting small defects, such as pitting and cracks, but it can be less effective on thicker pipes or pipes made of certain materials [90].
Electromagnetic Acoustic Transducer (EMAT)	- This method uses sound waves generated by electromagnetic fields to inspect the pipe wall [91,92]. - EMAT does not require a couplant like UT, which makes it useful for pipelines carrying gases or liquids that are incompatible with a couplant. - EMAT is still under development, but it has the potential to be a valuable tool for ILI [93].

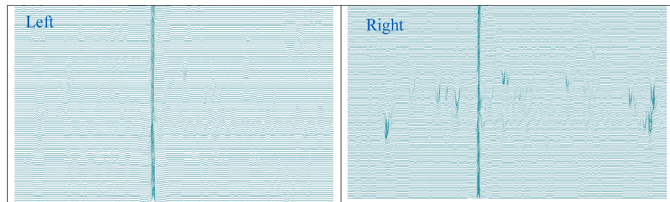


Fig. 8. Defect matching by comparing the raw magnetic signals.

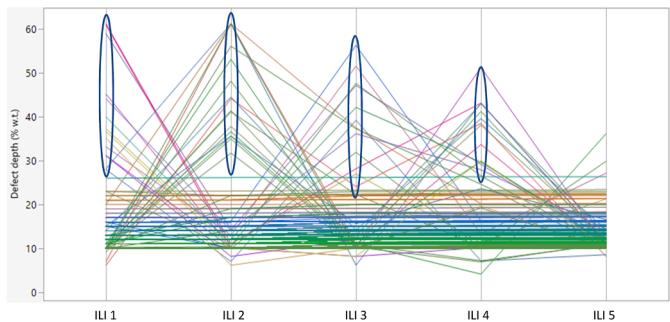


Fig. 9. Growth paths of individual anomalies determined by ILI measurements.

appropriate machine learning model for outlier detection in the pipeline ILI dataset, we propose a framework as shown in Fig. 7 based on the content and model types given in Table 2.

Selecting the right machine learning model for analysis of raw data is an important step in ensuring the accuracy and reliability of anomaly detection. Following the guideline shown in Fig. 7, we made a comparison among the ML models shown in Table 2 and chose Random

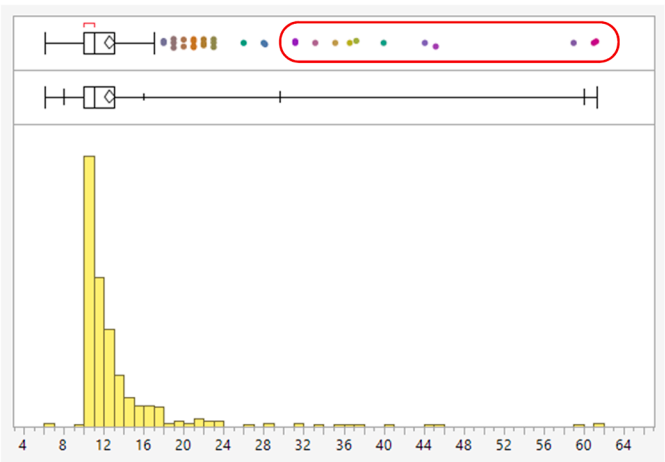


Fig. 10. ILI 1 data outliers.

Table 4
Quantiles Details for ILI 1 Data.

Quantiles		
100.0 %	maximum	61.25
99.5 %		59.995
97.5 %		29.580
90.0 %		16.00
75.0 %	Quartile	13.00
50.0 %	Median	11.00
25.0 %	Quartile	10.00
10.0 %		10.00
2.5 %		10.00
0.5 %		8.046
0.0 %	minimum	6.12

Table 5
Summary Statistics.

Summary Statistics	
Mean	12.620
Std Dev	5.7651
Std Err Mean	0.257824
Upper 95 % Mean	13.1267
Lower 95 % Mean	12.1136
N	500

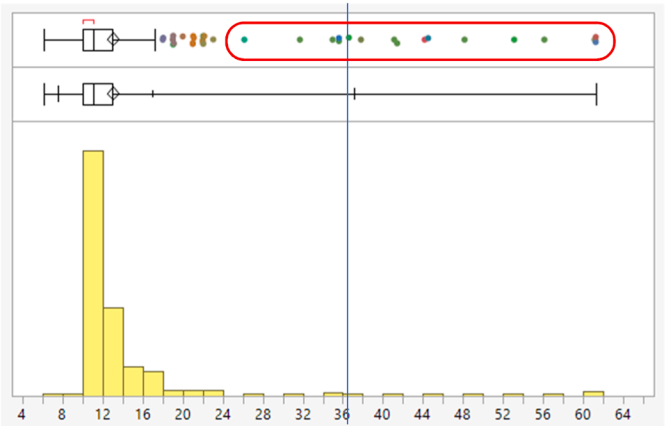


Fig. 11. ILI 2 Data Outliers.

Table 6
Quantiles Details for ILI 2 Data.

Quantiles		
100.0 %	maximum	61.26
99.5 %		61.2399
97.5 %		36.01
90.0 %		17.00
75.0 %	quartile	13.006
50.0 %	median	11.05
25.0 %	quartile	10.05
10.0 %		10.01
2.5 %		10.00
0.5 %		7.5857
0.0 %	minimum	6.15

Table 7
Summary Statistics.

Summary Statistics		
Mean		13.0443
Std Dev		7.0062
Std Err Mean		0.313328
Upper 95 % Mean		13.6599
Lower 95 % Mean		12.4287
N		500

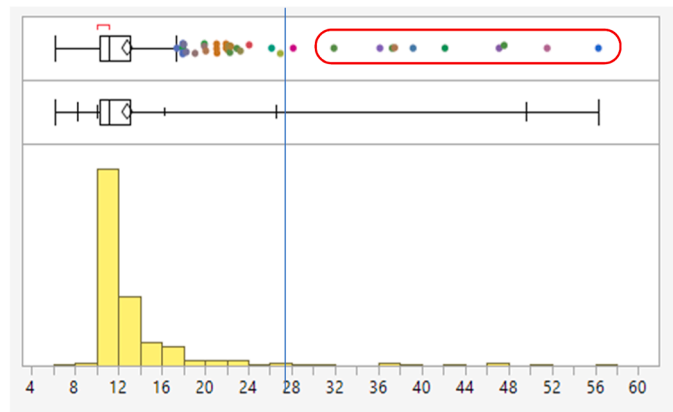


Fig. 12. ILI 3 Data Outliers.

Table 8
Quantiles Details for ILI 3 Data.

Quantiles		
100.0 %	maximum	56.31
99.5 %		49.575
97.5 %		26.54
90.0 %		16.259
75.0 %	quartile	13.086
50.0 %	median	11.16
25.0 %	quartile	10.27
10.0 %		10.086
2.5 %		10.05
0.5 %		8.16
0.0 %	minimum	6.16

Forest Regression to identify outliers in our case study. Random Forest Regression is well-suited for raw ILI data collected from oil pipelines due to several factors outlined below:

a. **Data Characteristics:** Random Forests can handle large datasets efficiently, making them suitable for large ILI datasets with multiple

Table 9
Summary Statistics.

Summary Statistics	
Mean	12.8633
Std Dev	5.23643
Std Err Mean	0.234180
Upper 95 % Mean	13.3234
Lower 95 % Mean	12.4032
N	500

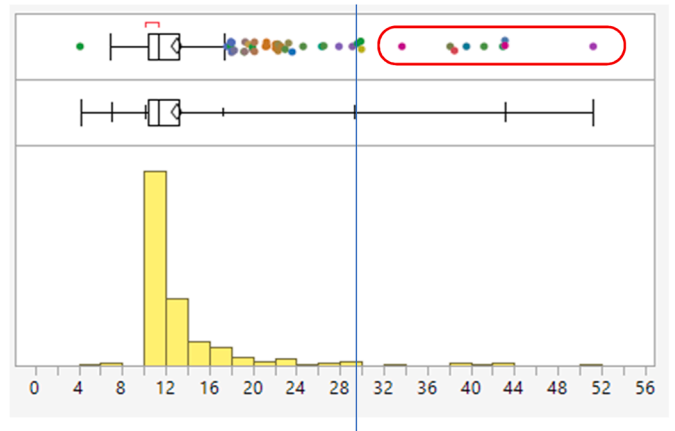


Fig. 13. ILI 4 Data Outliers.

Table 10
Quantiles Details for ILI 4 Data.

Quantiles		
100.0 %	maximum	51.26
99.5 %		43.15
97.5 %		29.366
90.0 %		17.16
75.0 %	quartile	13.157
50.0 %	median	11.296
25.0 %	quartile	10.39
10.0 %		10.146
2.5 %		10.05
0.5 %		7.016
0.0 %	minimum	4.12

Table 11
Summary Statistics.

Summary Statistics	
Mean	13.0286
Std Dev	5.10813
Std Err Mean	0.228443
Upper 95 % Mean	13.4774
Lower 95 % Mean	12.5798
N	500

features like defect size, length, depth, pipeline segment, location, and orientation.

b. **Nature of Data:** Random Forests can handle both numerical and categorical data, making them versatile for processing a combination of data types commonly found in ILI datasets.

c. **Distribution of Outliers:** Random Forests are robust against outliers and skewed data distributions, which is crucial for accurate outlier detection in pipeline data.

d. **Detection Goals:**

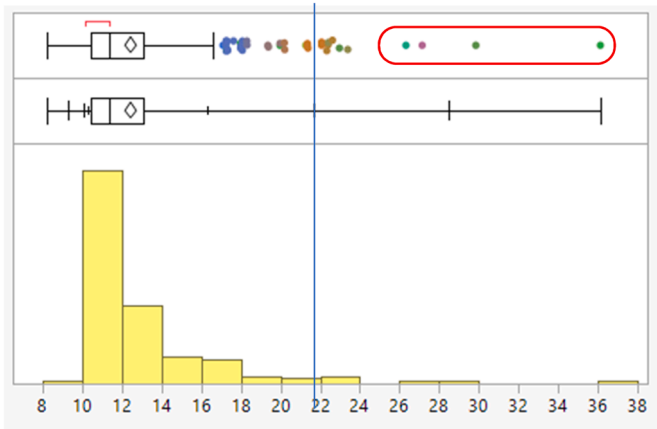


Fig. 14. ILI 5 Data Outliers.

Table 12
Quantiles Details for ILI 5 Data.

Quantiles			
100.0 %	maximum		36.15
99.5 %			28.491
97.5 %			21.708
90.0 %			16.3
75.0 %	quartile		13.07
50.0 %	median		11.356
25.0 %	quartile		10.43
10.0 %			10.25
2.5 %			10.087
0.5 %			9.302
0.0 %	minimum		8.16

Table 13
Summary Statistics.

Summary Statistics	
Mean	12.4352
Std Dev	3.0378
Std Err Mean	0.135855
Upper 95 % Mean	12.7021
Lower 95 % Mean	12.1683
N	500

Table 14
Benefits of Z-score to Identifying Outliers.

Benefits	Details
Normalization across scales	Allows for consistent comparison of data points regardless of original measurement units.
Quantitative assessment of deviation	Provides a numerical measure of how far a data point deviates from the mean, aiding in outlier identification.
Identification of extreme values	Flags extreme values based on threshold Z-scores, facilitating targeted outlier detection.
Statistical robustness	Based on mean and standard deviation, suitable for various data distributions, ensuring robust outlier detection.
Integration with statistical tests	Enhances outlier detection when integrated with statistical models and tests, improving data analysis accuracy.

- **Sensitivity:** Random Forests are less sensitive to outliers compared to models like Z-score and Mahalanobis Distance, making them suitable for outlier detection tasks.

- **Specificity:** If minimizing false positives and accurately identifying true outliers is essential, Random Forests provide a good balance between sensitivity and specificity.

e. **Computational Resources:** While Random Forests can be computationally intensive, they are generally more efficient compared to deep learning models, especially when dealing with large datasets.

f. **Performance Evaluation:** Random Forests can be evaluated using various metrics like precision, recall, F1-score, and ROC curve analysis, allowing for comprehensive performance assessment in outlier detection tasks.

5. Data collection and analysis

To collect pipeline condition inspection data using in-line inspection, several probes can be used. Each type has its strengths and weaknesses. Table 3 shows some of the most common types of probes used for pipeline assessment using in-line inspection.

Current approaches to matching an anomaly in the context of anomaly detection and analysis can be generally categorized into two main types:

- Signal comparison and
- Anomaly size and location (data derived from the signal) comparison.

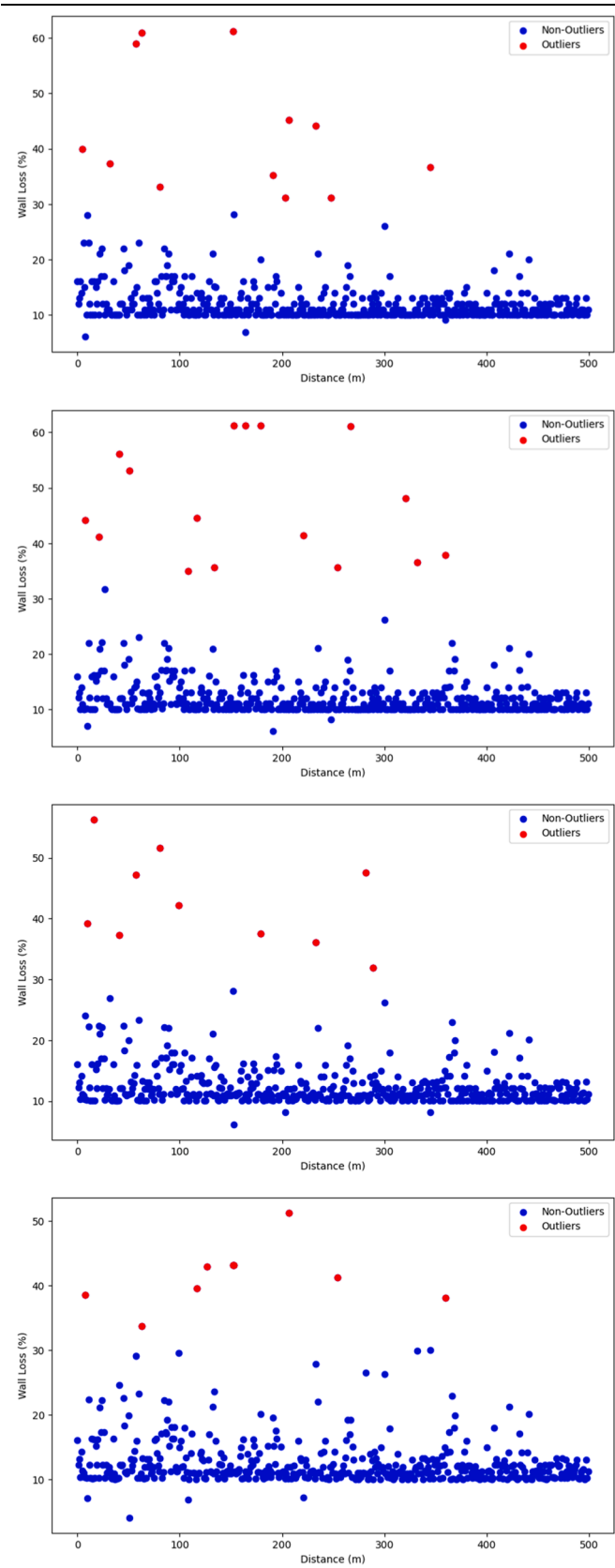
Fig. 8 illustrates the signal comparison approach, wherein raw signal data, particularly magnetic data [83,84], is visually represented, where the data shown is collected from two different inspection runs. By comparing the two graphs, i.e., the left and right images in Fig. 8, changes in defect size become apparent, enabling the observation of variations in the inspected material. While automatic detection is feasible, as demonstrated by Spencer and Tetreault [83], its practical application is typically constrained to scenarios where the defect density is low in a given area.

The signal comparison method is favoured for its enhanced accuracy compared to the second approach, as it circumvents the uncertainties inherent in algorithms that convert signals into defect sizes. While the signal comparison and defect size and location comparison methods each have its own merits, achieving unequivocal identification of a particular defect from one inspection to another remains a challenging task. This limitation underscores the complexity and dynamic nature of defect analysis, prompting ongoing efforts to enhance matching techniques for more accurate and reliable results. Researchers and industry professionals continue to explore innovative approaches to addressing these challenges and improve the overall effectiveness of defect detection. Achieving a definitive identification of a specific defect from one inspection to another is not directly feasible. It requires running defect-matching algorithms.

In this work, the ILI data was collected from a pipeline company in the Middle East for a pipeline commissioned in 1978. The data includes a wide range of features crucial for pipeline integrity assessment, such as pipeline segment distance (km), wall losses (%), anomaly length (mm), width (mm), corrosion growth rate and year of data collection. The ILI data was exported from ePipe to MS Excel spreadsheet with a clear identification of pipeline location, length, and wall losses. The ILI probes utilized to collect data for this selected oil pipeline is UT thickness measurement and MFL signal, where UT thickness was used for any wall losses and MFL for cracks. We analyzed wall losses to show data variations and their impact on pipeline condition assessment.

Out of 42,000 data samples, 500 data samples were selected randomly from different segments of the pipeline for the analysis. As seen in Fig. 9, the raw data indicates some large variations between different ILI measurements, collected over 22 years. The first inspection (ILI 1) was carried out in 2000, followed by 2005 (ILI 2), 2010 (ILI 3), 2015 (ILI 4) and 2022 (ILI 5). Such variations could happen for a few

Table 15
Outliers in Raw ILI data, Detected by Z-score Method.



The number of outliers identified by Z-score method for ILI 1 data is 12 which is the same as found by other methods.

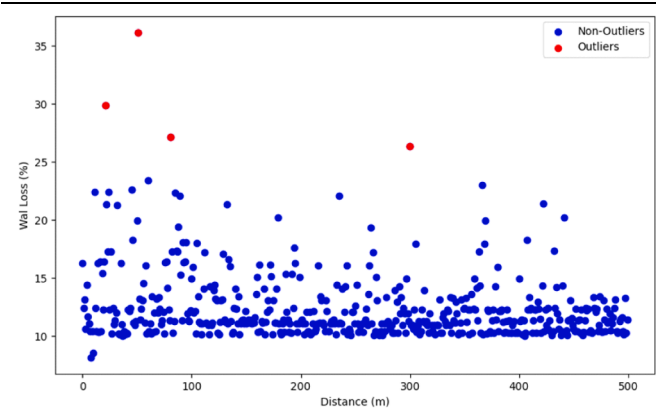
The outliers detected using Z-score method for ILI 2 data are 16, which is well aligned with the findings by other methods.

The Z-score method was employed to identify outliers in ILI 3 data. It is noteworthy that 10 outliers were found, demonstrating a perfect correspondence with those detected by other methods.

This plot illustrates the identification of outliers in ILI 4 data using Z-score method, a total 8 outliers are identified, which is very close to those identified by other methods.

(continued on next page)

Table 15 (continued)



ILI 5 dataset shows 4 outliers by Z-score method, which is the same as the identified by other methods.

Table 16
Test of Interquartile Range Method for Identifying Outliers.

Determination of threshold	Number of Outliers				
	ILI 1	ILI 2	ILI 3	ILI 4	ILI 5
$Q3 + 1.5 \text{ IQR}$	46	35	38	48	39
$Q3 + 4 \text{ IQR}$	14	17	12	16	5
95th percentile + 3 IQR	12	17	10	9	4

Table 17
Approach to Choose a Threshold Value for Pipeline ILI Raw Data.

Step	Brief Details
Understand Data Distribution	Begin by analyzing the distribution of your ILI data. Plot histograms or density plots to visualize the distribution. Understanding the data's central tendency and dispersion helps in setting an initial threshold.
Consider Statistical Methods	Statistical methods like Z-score or percentile-based approaches can be used to determine thresholds. Z-score identifies outliers based on how many standard deviations they are from the mean, while percentiles set thresholds based on data distribution.
Evaluate Impact	Consider the impact of setting different thresholds. A lower threshold may detect more outliers but could also increase false positives. Balancing sensitivity and specificity is essential.
Validation and Testing	Validate the chosen threshold using historical data or simulated scenarios if available. Testing the threshold helps assess its effectiveness in identifying true anomalies while minimizing false alarms.

Table 18
Overview of Random Forest Regression for Outliers' Identification.

Aspect	Description
Training Data	Historical ILI data includes pressure, temperature, flow rate, corrosion levels (anomaly values), etc.
Model Training	Ensemble decision trees trained on random subsets of data and features to reduce overfitting and improve generalization.
Prediction	Predicts normal behavior of ILI data based on features; data points with significant deviations from predictions are flagged as potential outliers.
Anomaly Detection	Analyze residuals (differences between actual and predicted values) to identify data points with unusually large residuals, indicating potential anomalies or outliers.
Feature Importance Analysis	It provides insights into which features are most influential in predicting normal ILI data behavior, aiding in understanding potential causes of outliers.
Thresholding	Sets thresholds based on residuals or other anomaly detection metrics to refine outlier detection and improve precision in identifying outliers.

reasons like probe selection, probe orientation, the person who collected the data, probe calibration, data transfer, and other reasons.

6. Statistical analysis of ILI data

ILI 1 data is analyzed by using the distribution method to identify outliers. This method identified a total 12 outliers (red circle) in ILI 1 raw data (Fig. 10). The distribution method is a statistical approach that involves analyzing the distribution of the data within a dataset to identify any outliers. The ILI data can be visualized using histograms, Q-Q plots, and box plots to identify data points that deviate significantly from the expected value.

From Box plot, it is observed that the data points with values above 28.15 may be considered as outliers in raw ILI 1 data. Quantiles details for ILI 1 data are shown in Table 4, where the 97.5th percentile is 29.580 which represents a defect depth in percentage of the pipe wall thickness (% w.t.). Therefore, any values above 29.580 (percentage) are considered outliers.

Table 5 shows summary statistics of ILI 1 raw data. This table shows the mean, standard deviation, standard error mean, and both upper and lower bounds of the mean of ILI 1 data at a 95 % confidence interval.

The analysis results for ILI 2 data are illustrated in Fig. 11 with outliers indicated within the red circle. Table 6 presents detailed quantiles information for the ILI 2 inspection data, where data points exceeding 26.16 are considered outliers. The 97.5th percentile is 36.01 and there are a total of 13 data points with values larger than 36.01. Specifically, the ILI 2 raw data reveals the presence of 17 outliers.

Table 7 shows summary statistics details for ILI 2 raw data.

Analysis results for ILI 3 data are presented in Fig. 12, where outliers are highlighted within the red circle. A total of 10 outliers are observed in ILI 3 dataset. Table 8 shows detailed information on the quantiles for ILI 3 inspection data. The 97.5th percentile is 26.54. If this value is set as a threshold, there are 11 outliers identified in ILI 3 data.

Table 9 shows summary statistics details for ILI 3 raw data.

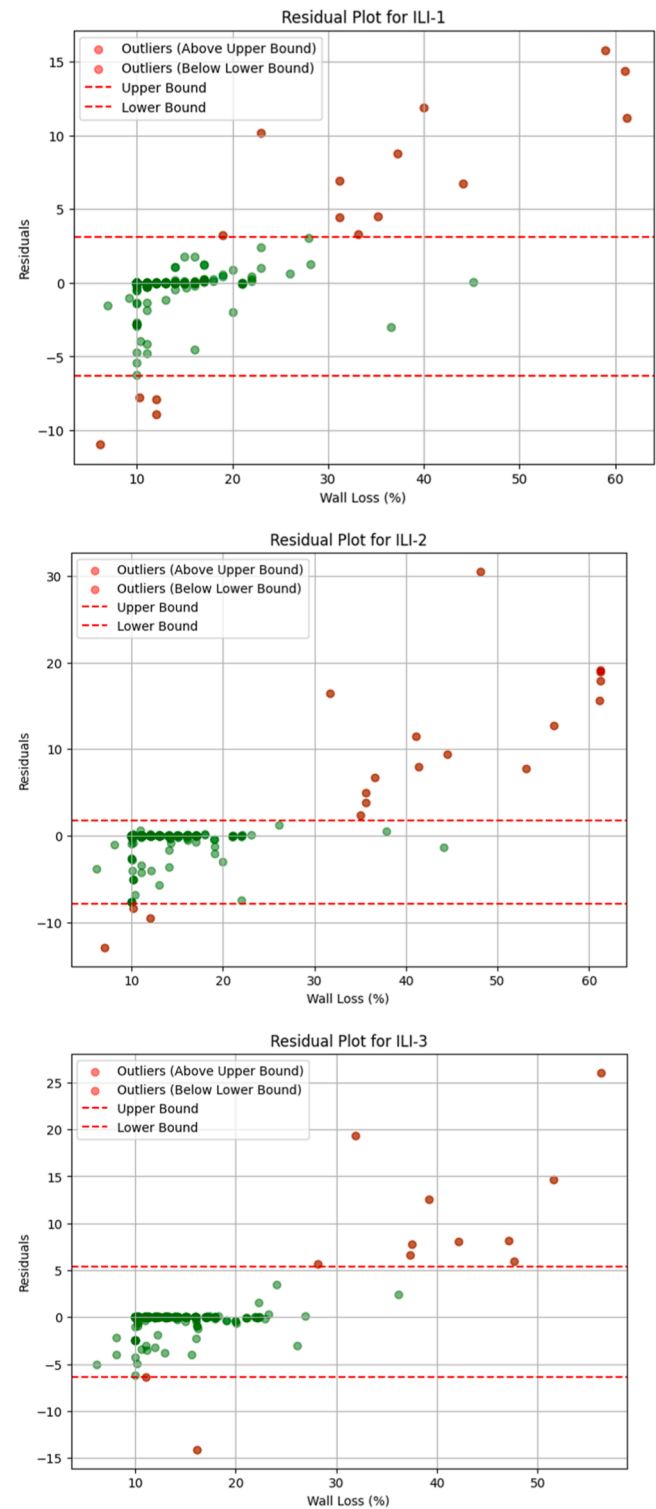
Fig. 13 shows analysis results for ILI 4 data, with outliers shown inside the red circle where 9 outliers are identified. Table 10 gives quantiles details for ILI 4 inspection data. The 97.5th percentile is 29.366. If it is chosen as the threshold, there could be a total of 12 outliers. Table 11 shows summary statistics details for ILI 4 raw data.

Fig. 14 shows analysis results for ILI 5 data with outliers shown inside the red circle. There are 4 outliers identified in ILI 5. Table 12 presents quantiles details of ILI 5 inspection data, where the 97.5th percentile is 21.708. There are a total of 12 data values larger than 21.708. Table 13 shows summary statistics details for ILI 5 raw data.

6.1. Z-score method

Z-score is a commonly used technique to identify outliers in raw

Table 19
Outliers in Raw ILI Data, Detected by Random Forest Regression.



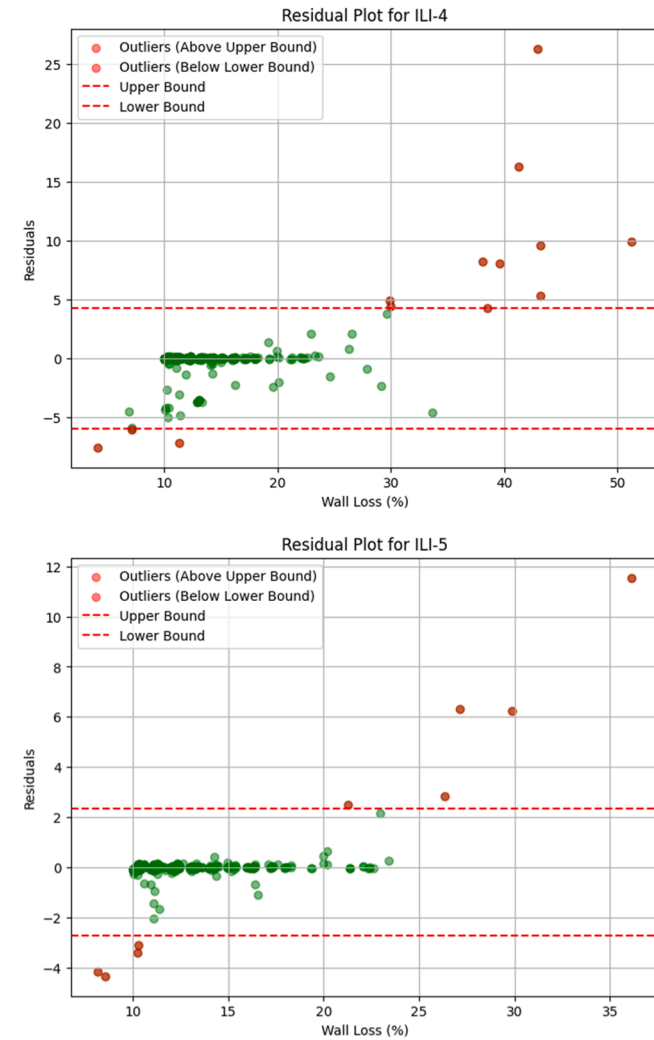
A total of 500 data samples were selected for this model analysis. This plot shows the total number of outliers for ILI 1 data, identified by Random Forest Regression. There are 12 outliers with values larger than the Upper bound. This is the same as identified by the distribution and Z-score methods (Section 6).

For ILI 2 data, a total of 15 outliers above the Upper bound were identified, which is approaching the number of outliers observed using Box plot where it is 17. Using Z-score method (Section 6.1), 16 outliers were identified.

This plot shows outliers identified in ILI 3 dataset. The analysis shows a total of 10 outliers above the Upper bound were pinpointed, which are of the same as identified by the distribution and Z-score methods.

(continued on next page)

Table 19 (continued)



This plot presents the outliers identified within the ILI 4 raw dataset. A total of 10 outliers above the Upper bound are found, which is 9 found by the distribution-based method and 8 found by Z-score method.

The outliers above the Upper bound detected within ILI 5 raw dataset are 5. This number closely reflects the total outliers given by the distribution and Z-score methods.

data. The Z-score, also known as the standard score or Z-value, measures the number of standard deviations by which a data point deviates from the dataset's mean [94]. The Z-score is calculated by

$$z = (x - \mu) / \sigma$$

where, z = Z-score, x is the value being evaluated, μ is the mean of the dataset, and σ is the standard deviation.

Table 14 shows a few benefits of using Z-score for identifying outliers in raw ILI data.

The outliers identified by Z-score method are shown in Table 15. The threshold value selected is ± 3 .

6.2. Test of interquartile range method

The interquartile range method is a popular method applied to detection of outliers in a dataset. It is first to determine the first quartile (Q1), the median, and the third quartile (Q3). The second step is to calculate interquartile range (IQR), $IQR = Q3 - Q1$. The third step is to calculate the upper fence (bound) which is equal to $Q3 + (1.5 \times IQR)$ and the lower fence (bound) which equals $Q1 - (1.5 \times IQR)$. The fourth step is to identify the outliers, that is, all values falling outside the fences are marked as outliers.

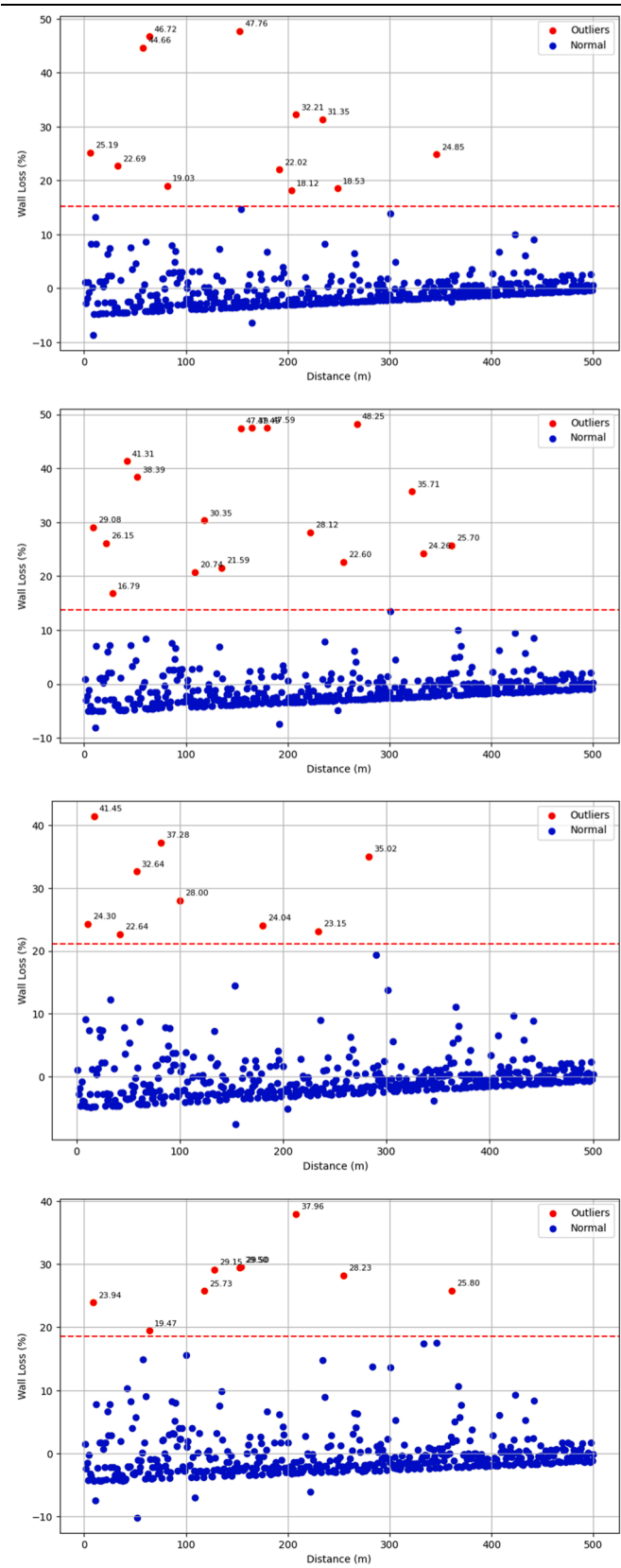
Applying the conventional interquartile range method to outlier detection for ILI 1 to ILI 5 datasets results in the detection results shown in Table 16. Clearly, the number of outliers detected is significantly

larger than other methods as discussed in the above and Section 7. If the upper fence is calculated using $Q3 + 4 \times IQR$, the number of outliers is still higher than the other methods. The reason why the conventional interquartile range method is not effective in the detection of outliers for the datasets is that the ILI measurement values in each dataset are highly left skewed. This is expressed by the histograms as shown in Figs. 11 to 14 in this section. Finally, we find a way to determine the upper fence by using the 95th percentile + $3 \times IQR$. Based on the new upper fences calculated, the detected outliers for each ILI dataset are closely aligned with those identified by using the above methods and machine learning models given in Section 7. Hence, we recommend using the 95th percentile + $3 \times IQR$ to calculate the upper fence when detecting outliers for the ILI data for the pipeline under this study.

7. Machine learning models for ILI data analysis

The application of machine learning (ML) models has become essential in the field of pipeline integrity management for the analysis of ILI raw data. However, ensuring the reliability and accuracy of these models requires rigorous testing and analysis, particularly in identifying outliers. The presence of outliers, anomalies, or unforeseen data points can have a substantial influence on the efficacy and reliability of ML models, potentially resulting in inaccurate findings or compromised safety evaluations. This section discusses testing of a few models on raw ILI data to identify outliers.

Table 20
Outliers in Raw ILI Data, Detected by Liner Regression Analysis.



The analysis reveals a total of 12 outliers, consistent with the findings derived from both the distribution and Z-score methods.

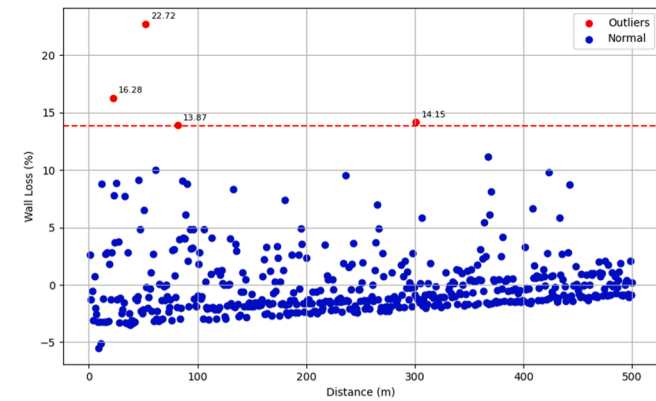
The total number of outliers detected for ILI 2 data is 17. This result is exactly aligned with the outcome from the distribution method, and approaching the findings by Z-score, and Random Forest Regression approaches.

The result for ILI 3 data analysis reveals 9 outliers in raw ILI data, which closely matches the findings from the distribution, Z-score, and Random Forest Model methods.

ILI 4 data showcases 9 outliers, aligning closely with the outcomes observed through the distribution and Z-score methods.

(continued on next page)

Table 20 (continued)



ILI 5 data shows a total of 4 outliers. This result is in close agreement with the findings using the distribution, Z-score, and Random Forest Model methods.

7.1. Random forest regression

One of the machine learning algorithms commonly employed to detect outliers in ILI data is Random Forest Regression. This approach is designed to predict a continuous variable by utilizing input data [95]. The ensemble approach is a technique that integrates many decision trees to provide predictions. Random Forest Regression offers valuable insights into the significance of features, aiding in the identification of the most influential features in the prediction process [96]. Selecting a threshold value is an important step in outlier detection for pipeline ILI raw data. Refer to Table 17 for the guideline in selection of the threshold. The threshold value for the Random Forest Regression model is ± 3 .

An overview of the utilization of Random Forest Regression for the identification of outliers in ILI data is presented in Table 18.

The analysis results to identify outliers in raw ILI data using the Random Forest Regression model is shown in Table 19, covering the outliers for ILI 1 to ILI 5 data.

7.2. Linear regression analysis

Linear regression analysis assumes a linear relationship between the variables, making it a valuable tool for prediction and inference. For linear regression analysis, along with the dependent variable (wall loss), we considered age of the pipeline as an independent variable, that is the number of years of the pipeline in service.

The threshold value for linear regression analysis is selected ± 3 . In linear regression, the threshold typically refers to a value used to categorize or filter the data based on the prediction using the independent variables. However, in the context of wall loss determination, the threshold is determined as follows:

Threshold in Wall Loss Assessment: This could involve setting a critical wall loss percentage, beyond which the pipeline is considered at risk of failure.

Statistical Threshold: In the analysis, the threshold could also refer to a decision rule based on the regression output. For instance, if the regression model predicts a certain level of wall loss based on the independent variables, a threshold might be set to trigger maintenance actions when the predicted wall loss exceeds a specific percentage.

The results of linear regression analysis for outlier detection of ILI 1 to ILI 5 datasets are presented in Table 20.

7.3. K-Nearest neighbors (KNN) model

The KNN model is widely utilised in machine learning for both classification and regression tasks. The capabilities of this model also encompass outlier detection, rendering it a versatile instrument for data analysis, particularly within the realm of pipeline ILI raw data.

The K-nearest neighbors (KNN) algorithm functions by leveraging the concept of similarity, wherein data points are classified or assessed according to their proximity to adjacent points within the feature space. The utilization of KNN in outlier detection enables the identification of data points that exhibit substantial deviation from their neighboring data points, hence suggesting the presence of potential anomalies or outliers.

KNN's versatility in managing data with a large number of dimensions and its capacity to adjust to different data distributions make it a powerful tool for identifying outliers in intricate datasets like ILI raw data. Using the KNN method, the outlier detection results for ILI 1 to ILI 5 data are shown in Table 21.

8. Conclusion

Data quality issues, such as inconsistencies, missing values, missed defect matching, long distance mismatching, girth weld mismatching and outliers, can lead to inaccurate assessments and misinterpretation of inspection results. These issues can have severe consequences, potentially leading to missed anomaly and misinterpretation of defects resulting in safety hazards and operational interruptions. Analyzing and identifying outliers in raw ILI data provides great value addition to oil and gas pipeline integrity management. This research work has examined machine learning models and compared these models with statistical analysis technique in detecting outliers. Our findings reveal a significant overlap in the results obtained from both statistical analysis and machine learning approaches in detecting outliers of ILI data. Machine learning models, despite their complexity and need for substantial computational resources, demonstrated a high level of accuracy in identifying outliers, matching the performance of traditional statistical methods. This convergence of results suggests that machine learning models can serve as a robust alternative or complement to statistical techniques, especially in scenarios where the data is vast and complex. Our study also confirms that it is critical to determine a proper threshold in outlier detection for a given dataset in order for a method to be effective in the detection. This depends on the nature and characteristics of the data in analysis. For example, the conventional interquartile range method is found not effective to detect the outliers for the ILI datasets under study, a new modification of the method to calculate the upper fence by the 95th percentile + 3 IQR is proposed and it is tested having the same performance as other methods utilized.

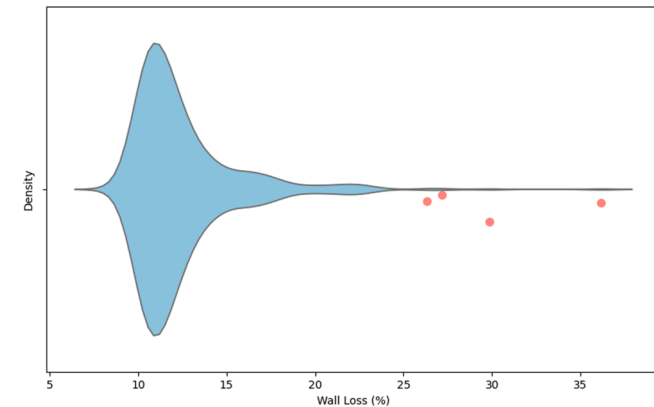
To test multiple approaches in this study is to propose a method for outlier detection for achieving robust results. On one side, one should be conservative to take all the detected outliers for checking and verification to make sure whether they are the true outliers or false measurements. On the other side, however, we need to make sure that these required checking and verifications are not to cause much additional work to sort out those which are not the true outliers. The choice of these

Table 21
Outliers in Raw ILI Data, Detected by KNN Technique.

	The implementation of the K-nearest neighbors (KNN) algorithm for outlier detection in ILI 1 data is illustrated. The analysis shows the presence of a total of 12 larger outliers, exactly the same as the number of outliers identified by other techniques.
	The KNN model detects 15 larger outliers in ILI 2 data. This result is consistent with the findings obtained from other techniques discussed above.
	This plot shows the analysis results by utilizing the KNN algorithm to the ILI 3 raw data, showing 9 larger outliers, which aligns with the identifications made by other models and techniques.
	The analysis outcome using the KNN algorithm on the ILI 4 raw dataset reveals 7 larger outliers, which are very close to the total number of outliers detected by other techniques.

(continued on next page)

Table 21 (continued)



This graph depicts the analysis results of employing the KNN algorithm on ILI 5 raw dataset, revealing 4 larger outliers, which is the same as the number of outliers detected by other methods.

approaches should be guided by the specific requirements of the analysis, the nature of the dataset, and the available resources. For further study, one may explore hybrid approaches that leverage the strengths of both machine learning and statistical methods, potentially leading to even more robust and reliable outlier detection frameworks. This integration could further enhance data quality, contributing to safer and more efficient infrastructure management.

9. Future directions

Regarding the future study, one of the directions is the utilization of machine learning models tailored to the unique characteristics of pipeline ILI raw data. Given the high-dimensional and often noisy nature of the raw ILI data, there is potential to develop more sophisticated models that can accurately distinguish between true anomalies and noise. Further research could focus on the integration of hybrid models that combine machine learning techniques with traditional statistical methods. This approach could enhance the robustness and interpretability of outlier detection systems. For instance, machine learning models could be employed to preprocess data, reducing noise and isolating potential outliers, which can then be further validated using statistical analysis.

Comparative studies between machine learning models and traditional statistical methods should be expanded to assess their performance across various pipeline conditions and data types. These comparisons could help identify the strengths and limitations of each approach, guiding the selection of appropriate techniques for specific pipeline integrity challenges.

Other research works may be related to the following aspects:

a. Automated Outlier Detection Algorithms Development

- Design and implement automatic outlier detection algorithms customized for raw ILI data analysis.
- Utilize machine learning algorithms and statistical approaches to detect anomalies in real-time or near-real-time.

b. Integration with Pipeline Monitoring Platforms

- Incorporate outlier detection capabilities into existing pipeline monitoring platforms.
- Enable ongoing surveillance of ILI data streams, identifying anomalies as they arise.

c. Enhanced Data Visualization and Reporting

- Improve data visualization techniques and reporting mechanisms to highlight outliers effectively.
- Develop interactive dashboards, anomaly heatmaps, and trend analysis tools.

d. Predictive Analytics for Outlier Prediction

- Construct predictive analytics models that utilize historical trends in ILI data, environmental conditions, and operational parameters.

- Forecast potential outliers to enable proactive mitigation methods.

e. Risk-Based Outlier Prioritization

- Implement risk-based outlier prioritization frameworks.
- Prioritize the removal of outliers based on the severity of potential integrity issues.
- Enhance maintenance efforts and mitigate operational hazards by focusing on high-risk outliers.

f. Continuous Model Refinement and Validation

- Continuously improve and validate outlier detection models.
- Incorporate feedback from field inspections, maintenance activities, and historical outcomes related to outlier removal.
- Use an iterative technique to ensure the ongoing effectiveness and adaptability of outlier identification algorithms in response to changing data patterns and abnormalities.

Interdisciplinary collaboration between data scientists, pipeline engineers, and industry experts will be crucial in advancing the practical implementation of machine learning-based outlier detection systems. Such collaborations could lead to the development of standardized protocols and best practices, facilitating the widespread adoption of these technologies in the pipeline industry.

To summarise, the progress in outlier detection technologies, collaborative industry initiatives, and risk-based outlier management approaches will be essential in helping pipeline owners examine raw ILI data for outliers and highlight the significance of removing outliers for the overall integrity and operational excellence of the pipeline.

CRedit authorship contribution statement

Muhammad Hussain: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tieling Zhang:** Writing – original draft, Conceptualization, Methodology, Formal analysis, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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